

# WeiQi (Rocky) Wang

Robotics Engineer · UCLA Ph.D. Graduate

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## EDUCATION

**University of California, Los Angeles** — Ph.D., Mechanical Engineering (Robotics) · September 2026

*Dissertation: Embodiment-Aware Task Realization for Human–Robot Coexistence*

**Johns Hopkins University** — M.S., Mechanical Engineering (Robotics) · May 2017

**University of Miami** — B.S., Mechanical Engineering; Minor in Mathematics · May 2015

## RESEARCH EXPERIENCE

**UCLA · Center for Vision, Cognition, Learning, and Autonomy | Graduate Student Researcher**  
September 2020–Present

### SUN: Agentic Robot Policy Learning with Persistent Task Programs

- Built Kuafu, an agentic robot-learning harness that turns object videos and language instructions into learned manipulation policies using shared optimal-control and reinforcement-learning objectives.
- Developed an agent-driven workflow for task decomposition, program verification, diagnosis, and repair, achieving **95.6% formation-and-verification success** (43/45 runs).
- Achieved **82.03% mean success across nine simulated multi-stage tasks** with behavior cloning and residual RL; exceeded evaluated LLM reward-design and zero-shot language-guided planning methods by 69.18 and 72.03 percentage points, respectively.
- Built a web-based teleoperation application for collecting human demonstrations in simulation with AgileX PIKA task-space controls.
- Automated successful-demonstration collection at **10.57× human teleoperation throughput** after training; including controller preparation, broke even at 35 demonstrations on cost and 513 on time per task on average.
- Trained RGB, point-cloud, and multi-task VLA policies (SmoVLA, WALL-X,  $\pi_{0.5}$ ). DP3 reached **46.02% mean success** on nine simulated tasks, 23.6 percentage points above the matched baseline.
- Developed the DP3 deployment pipeline for **zero-shot sim-to-real transfer on Franka FR3 and Kinova Gen3** across three robot–gripper configurations, without real-world demonstrations or fine-tuning.

### Scene Rearrangement for Human–Robot Co-Activity

- Led a scene-rearrangement system using hierarchical ASA/CMA-ES optimization, robot footprint and reachability models, learned spatial relationships, and ConceptNet semantics.
- Improved accessible space by **14%**, with **30% more reachable objects** in the published scene evaluation.

### Collaborative Motion Planning and Tool Use

- Contributed to optimization-based mobile manipulation using Virtual Kinematic Chains. Developed trajectory optimization for simulated Baxter tool-use motions and verified the formulation in MATLAB.

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## INDUSTRY EXPERIENCE

### OffWorld, Inc. | Software Intern | Pasadena, CA

June–September 2023; June–September 2021

- Developed a Python computational-geometry simulation framework for prototype autonomous underground mining robots, modeling environmental changes caused by robot sawing actions.
- Modeled tool-mounted serial-manipulator dynamics and enabled task-specific force-feedback control. Developed sampling- and optimization-based dual-arm motion planning, with 100% planning success in the evaluated scenarios.
- Built ROS/ROS 2, Gazebo, and RViz testbeds for multi-robot coordination and navigation, and procedurally generated Webots scenarios for varied test conditions.

### CloudMinds / INNFOSS Technology Ltd. | Algorithm Engineer | Beijing, China

December 2017–September 2019

- Led development and maintenance of ROS control software in C++ for the XR1 dual-arm mobile humanoid. Implemented wearable-IMU teleoperation of a 34-degree-of-freedom robot.
- Developed dynamic parameter identification for mobile dual-arm robots; applied the models to dynamic compensation, collision detection, and impedance/admittance control.
- Tuned position, velocity, and current control loops for joint actuators. Engineered quadruped walking and trotting control in C++ with six-degree-of-freedom body-pose regulation.

## EARLIER RESEARCH & TEACHING

### Johns Hopkins University · Laboratory for Computational Sensing + Robotics

Research Assistant | June 2016–September 2017

- Evaluated fiber Bragg grating shape sensors; built a MATLAB interface for real-time needle reconstruction and integrated low-level control. Contributed to optimal-control virtual fixtures for collaborative ultrasound imaging.
- Led a haptic femur-drilling simulator in CHAI3D, integrating CT-derived anatomical models and haptic feedback. Implemented model-based gravity compensation for the da Vinci Research Kit master arm.

**UCLA · Instructor & Teaching Fellow, Mathematics for Life Scientists** — 18 quarters; nominated for the Distinguished Teaching Award for Teaching Assistants.

## TECHNICAL SKILLS

**Languages & tools:** Python, C++, MATLAB, PyTorch, ROS/ROS 2, CasADi, Linux, Git, SolidWorks.

**Learning:** Imitation learning, behavior cloning, residual RL, visuomotor policies, VLA training, sim-to-real.

**Planning & control:** MPC, trajectory optimization, sampling-based planning, robot dynamics, system identification, impedance/admittance control, teleoperation.

**Perception:** 6D pose estimation, state estimation, sensor fusion, point-cloud registration, RGB-D calibration.

**Simulation & hardware:** Isaac Lab/Sim, MuJoCo, PyBullet, Gazebo, Webots, CoppeliaSim, CAN bus, UR5, Franka FR3, Kinova Gen3.

## SELECTED PUBLICATIONS

**Wang, W.**, et al. "SUN: Agentic Robot Policy Learning with Persistent Task Programs." *Under Review*, 2026.

**Wang, W.**, et al. "Rearrange Indoor Scenes for Human–Robot Co-Activity." *IEEE ICRA*, 2023.

Zhang, Z., et al. "Understanding Physical Effects for Effective Tool-Use." *IEEE RA-L*, 2022.

Jiao, Z., et al. "Efficient Task Planning for Mobile Manipulation: A Virtual Kinematic Chain Perspective." *IEEE/RSJ IROS*, 2021.

Zahedi, E., et al. "Towards Skill Transfer via Learning-Based Guidance in Human–Robot Interaction: An Application to Orthopaedic Surgical Drilling Skill." *Journal of Intelligent & Robotic Systems*, 2020.